



**Nolan Brechtel (Project Manager), George Utz (Chief Scientist),
Cara Belew, Zachary Morrow, Rebecca Ueltschey
Department of Electrical and Computer Engineering
Mississippi State University**



- Approximately 25,000 people, including more than 10,000 children, die every day due to starvation [1]
- Agricultural imports in the U.S. exceed its exports by \$41 billion each year [2]
- 49% of U.S. crop fields struggle with poor health and resource concerns like nutrient imbalances [3]

- **ROGO Soil Sampling Robot**
- **Large, requires fuel to operate**
- **Expensive and is typically rented**
- **Agrobot Lala**
- **Requires lab area set up to test soil**

- Inexpensive**
 - Price is less than \$1,000
- Small and Compact**
 - Can easily be stored when out of use
- Solar Powered**
 - No need for gas or any other power source
- Self-Contained**
 - Can test soil without needing to go to a set lab

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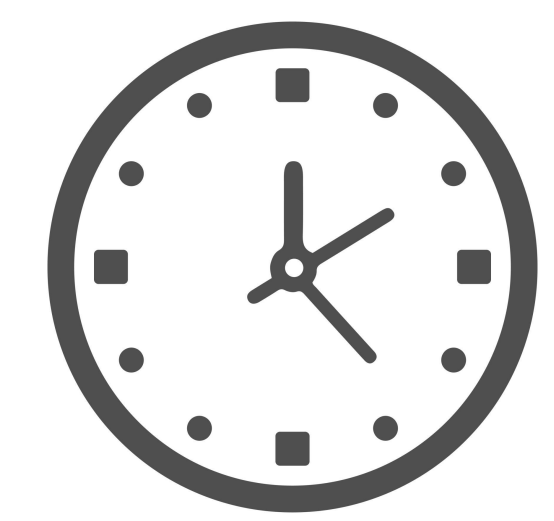
graph TD
    subgraph Navigation
        GPS
        Camera
    end
    subgraph PowerCharging [Power/Charging]
        VoltageRegulator[Voltage Regulator]
        SolarPanel[Solar Panel]
        Battery
        Power
    end
    subgraph ControlData [Control and Data]
        RaspberryPi[Raspberry Pi]
        Radio
        uController
    end
    subgraph SoilTesting [Soil Testing]
        pHTesting[pH Testing]
        MoistureSensing[Moisture Sensing]
        NPKTesting[NPK Testing]
    end
    subgraph DriveTrain [Drive Train]
        Motor
        MotorController[Motor Controller]
        Drill
        Probes
    end

    SolarPanel --> Battery
    Battery --> Power
    Power --> VoltageRegulator
    VoltageRegulator --> uController
    VoltageRegulator --> MotorController
    VoltageRegulator --> DriveTrain

    uController --> GPS
    uController --> Camera
    uController --> pHTesting
    uController --> MoistureSensing
    uController --> NPKTesting
    uController --> MotorController

    RaspberryPi <--> Radio
    Radio <--> uController
  
```

- **Customer Impact**
 - Allows for testing both more quickly and around the clock



- **Broader Impact**
 - Gives farmers the tools to make more nutritious food



- **Future Expansion**
 - Swappable sensor to test for air pollution, noise pollution, poison, etc



1. J. Holmes. "Losing 25,000 to Hunger Every Day." [un.org](#). Accessed: Sep. 9, 2026. [Online.] Available: [Losing 25,000 to Hunger Every Day | United Nations](#)
2. J. Kaufman. "Ag and Food Statistics Charting the Essentials." [ers.usda.gov](#). Accessed: Sep. 9, 2026. [Online.] Available: [Ag and Food Statistics: Charting the Essentials - Agricultural Trade | Economic Research Service](#)
3. A. B. Rosenberg and S. Wallander "Farmers Report Soil Related Resource Concerns on About Half of Soybean Wheat Cotton and Oat Fields." [ers.usda.gov](#). Accessed: Sep. 9, 2026. [Online.] Available: [Farmers Report Soil-Related Resource Concerns on About Half of Soybean, Wheat, Cotton, and Oat Fields | Economic Research Service](#)